

A Comparative Study of Random-Walk, Langevin, Barker, and Hamiltonian MCMC on PosteriorDB

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Overview of `sample_mcmc()`. All experiments in this paper are produced by a single driver, `sample_mcmc()`, whose purpose is to enforce a *common experimental protocol* across methods. Given (i) a target posterior π (specified through a log density and, when required, its gradient), (ii) a sampler choice (AMH, MALA, Barker, Bimodal Barker, HMC), and (iii) a tuning/adaptation configuration (initial scale, target acceptance probability, and either diagonal or full-matrix shape adaptation), `sample_mcmc()` runs M independent chains with an explicit warm-up period followed by a sampling period with *frozen* tuning parameters. This “adapt-then-freeze” design is essential: it isolates adaptation to a transient phase (warm-up), and it ensures that the retained draws are generated by a *time-homogeneous* Markov kernel with stationary distribution π [1]. For PosteriorDB targets (Section 4), we interface Stan models through BridgeStan, which provides efficient in-memory access to log densities, gradients, and (un)constraining transforms [2]. Reference draws and “ground truth” moments (used in running diagnostics) are taken from the PosteriorDB repository [3].

1 Introduction

Modern Bayesian workflows frequently require sampling from posterior distributions that are high-dimensional, strongly correlated, and often far from Gaussian. Markov chain Monte Carlo (MCMC) remains the principal tool for this task, but its practical success hinges on a trade-off that is both conceptual and operational: *efficiency* versus *robustness-to-tuning*. On the one hand, simple Metropolis–Hastings methods with symmetric random-walk proposals are mechanically reliable and broadly applicable, but they typically suffer from diffusive, random-walk exploration in moderate to high dimension. On the other hand, gradient-informed methods—such as the Metropolis-adjusted Langevin algorithm (MALA) and Hamiltonian Monte Carlo (HMC)—can yield dramatically improved mixing by aligning proposals with the local geometry of the target, but this improvement comes at the cost of additional structure (gradients, integrator choices) and increased sensitivity to numerical parameters (step sizes, mass matrices, trajectory lengths).

This tension is amplified by the fact that “good” tuning is rarely a static, model-independent choice. Even for classical families of targets, asymptotic scaling theory indicates that optimal efficiency occurs in narrow operating regions characterized by specific acceptance probabilities. For instance, diffusion-limit calculations suggest an optimal acceptance probability near 0.234 for the random-walk Metropolis algorithm on product targets [4], near 0.574 for MALA under similar asymptotics [5], and (in a separate asymptotic regime) near 0.651 for HMC when tuned via leapfrog step size [6]. These results are informative but also underscore a practical reality: the performance landscape can be steep, and departures from the tuned regime may cause rapid degradation. Consequently, *robustness*—understood as resilience to imperfect tuning and imperfect adaptation—is a first-class criterion, not merely an implementation detail.

A recent contribution in this direction is the *Barker proposal* of [7]. The method belongs to a broader class of *locally balanced* Metropolis–Hastings proposals, where the proposal mechanism itself is biased using first-order (gradient) information, but in a way designed to be less brittle than standard Langevin proposals. The Barker proposal is inspired by the classical Barker acceptance function [8] through a specific balancing choice, and it is motivated explicitly by the question: not only “how fast is the method when optimally tuned?”, but also “how badly does it behave when tuning is imperfect?” [7]. Closely related work shows that modifying the *auxiliary noise distribution* within the Barker construction can further improve performance, sometimes substantially, on representative targets [9].

The goal of this paper is therefore empirical and comparative: we implement a compact family of samplers—from adaptive random-walk Metropolis (AMH) to adaptive MALA (AMALA), Barker-type proposals (with both Gaussian and bimodal auxiliary noise), and HMC—and benchmark them under a unified interface and diagnostic pipeline. The emphasis is methodological rather than purely computational: we report standard end-of-run convergence and efficiency diagnostics (e.g. $\hat{R}$, effective sample size, Monte Carlo standard error, runtime), and we also compute running-time performance curves that reveal *how quickly* each method approaches correct

posterior moments. Crucially, we do not restrict attention to a single “best tuned” configuration. Instead, our framework is designed to evaluate both: (i) efficiency when the adaptation succeeds, and (ii) robustness when tuning and geometry are more challenging.

2 Samplers

2.1 Common notation and the Metropolis–Hastings backbone

Let $\pi : \mathbb{R}^d \rightarrow (0, \infty)$ denote the target density in *unconstrained* coordinates, known up to a normalizing constant. (For Stan models, π is the posterior density after applying Stan’s unconstraining transform; see Section 4.) A generic Metropolis–Hastings (MH) transition from $x \in \mathbb{R}^d$ proceeds by:

1. proposing $y \sim q(\cdot | x)$, where $q(y | x)$ is a proposal density on $\mathbb{R}^d$,
2. accepting y with probability

$$\alpha_{\text{MH}}(x, y) = \min \left\{ 1, \frac{\pi(y) q(x | y)}{\pi(x) q(y | x)} \right\}, \quad (1)$$

and otherwise staying at x .

If the proposal is symmetric, $q(y | x) = q(x | y)$, then (1) reduces to $\alpha_{\text{MH}}(x, y) = \min\{1, \pi(y)/\pi(x)\}$ [10–12].

Across the samplers considered here, we separate *scale* and *shape*:

- A scale parameter $s > 0$ controls the overall magnitude of proposed moves.
- A shape matrix $\Sigma \in \mathbb{R}^{d \times d}$ (symmetric positive definite) controls anisotropy and correlation. We write $\Sigma = LL^\top$ with L lower triangular.

Our implementation exposes two families of shape adaptation: a diagonal (variance-only) estimate and a full (covariance) estimate, motivated by standard adaptive MCMC practice [1, 13].

2.2 Adaptive Metropolis–Hastings (AMH)

Idea. AMH is a random-walk Metropolis method whose proposal covariance is adapted during warm-up to match the empirical covariance of the target [1, 13]. It serves as a robust baseline: it requires only evaluations of $\log \pi(x)$ (no gradients), but it is vulnerable to random-walk scaling in high dimension [4].

Proposal. Given the current state $x_t \in \mathbb{R}^d$, propose

$$y = x_t + s_t L_t z, \quad z \sim \mathcal{N}(0, I_d), \quad L_t L_t^\top = \Sigma_t. \quad (2)$$

Equivalently, $y | x_t \sim \mathcal{N}(x_t, s_t^2 \Sigma_t)$, which is symmetric. The acceptance probability is therefore $\alpha_{\text{MH}}(x_t, y) = \min\{1, \pi(y)/\pi(x_t)\}$.

Adaptation (warm-up only). During warm-up, we update:

- *Scale adaptation:* update $\log s_t$ by a Robbins–Monro scheme targeting a prescribed acceptance probability $\tau^* \in (0, 1)$:

$$\log s_{t+1} = \log s_t + \gamma_t (\alpha_t - \tau^*), \quad \gamma_t = c t^{-\kappa}, \quad \kappa \in (1/2, 1], \quad (3)$$

where α_t is the realized acceptance indicator (or acceptance probability) at iteration t . Such schemes are standard in adaptive MCMC [1, 14].

- *Shape adaptation:* set Σ_t to either (i) a diagonal matrix of running marginal variances (cheap, ignores correlations) or (ii) a full empirical covariance estimate (captures correlations but is costlier). In both cases we use a regularization $\Sigma_t \leftarrow \Sigma_t + \epsilon I_d$ with small $\epsilon > 0$ to ensure positive definiteness.

After warm-up, (s_t, Σ_t) are frozen at their final values, yielding a homogeneous Markov chain for which standard MCMC diagnostics apply directly.

2.3 Adaptive Metropolis-adjusted Langevin algorithm (MALA)

Idea. MALA modifies the random walk by adding a drift term proportional to the gradient $\nabla \log \pi(x)$, approximating a Langevin diffusion whose stationary density is π [15]. The resulting proposal is asymmetric and must be corrected by MH.

Preconditioned MALA proposal. Given $x_t \in \mathbb{R}^d$, propose

$$y = x_t + \frac{s_t^2}{2} \Sigma_t \nabla \log \pi(x_t) + s_t L_t z, \quad z \sim \mathcal{N}(0, I_d), \quad L_t L_t^\top = \Sigma_t. \quad (4)$$

Thus

$$y | x_t \sim \mathcal{N}\left(x_t + \frac{s_t^2}{2} \Sigma_t \nabla \log \pi(x_t), s_t^2 \Sigma_t\right),$$

so $q(\cdot | x_t)$ is Gaussian but not symmetric.

MH correction. The acceptance probability is (1) with the MALA Gaussian proposal. Writing $m(x) = x + \frac{s^2}{2} \Sigma \nabla \log \pi(x)$ and using s, Σ fixed (post warm-up) for readability,

$$\log \frac{q(x | y)}{q(y | x)} = -\frac{1}{2s^2} \|L^{-1}(x - m(y))\|^2 + \frac{1}{2s^2} \|L^{-1}(y - m(x))\|^2, \quad (5)$$

and therefore

$$\alpha_{\text{MALA}}(x, y) = \min \left\{ 1, \exp \left(\log \pi(y) - \log \pi(x) + \log \frac{q(x | y)}{q(y | x)} \right) \right\}. \quad (6)$$

MALA can substantially outperform random-walk MH on smooth targets, but its performance is sensitive to step size. Under classical product-target asymptotics, the optimal acceptance probability is about 0.574 [5].

Adaptation. Our MALA implementation uses the same warm-up strategy as AMH: a scale update of the form (3) and either diagonal or full covariance shape adaptation. As with AMH, adaptation is disabled after warm-up.

2.4 The Barker proposal (locally balanced MH)

Idea. The Barker proposal of (author?) [7] is a gradient-based Metropolis–Hastings method constructed within a *locally balanced* framework. Rather than using the Langevin drift to shift the mean of a Gaussian proposal (as in MALA), the Barker proposal biases the *direction* of a symmetric base step using a logistic weighting derived from a *balancing function*. The resulting proposal is designed to maintain high efficiency while improving robustness to imperfect step-size tuning [7].

Balancing function and proposal kernel. Let $\mu_\sigma(\cdot)$ be a symmetric base density on $\mathbb{R}^d$ for increments (in practice Gaussian with scale and shape). A balancing function $g : (0, \infty) \rightarrow (0, \infty)$ is required to satisfy

$$g(t) = t g(1/t), \quad (7)$$

which ensures local reversibility properties in the locally balanced construction. The *Barker* choice is

$$g(t) = \frac{t}{1+t}. \quad (8)$$

With this choice, (author?) [7] show that the resulting continuous-state proposal density can be written in closed form as

$$q_B(y | x) = \frac{2 \mu_\sigma(y - x)}{1 + \exp\{-\langle \nabla \log \pi(x), (y - x) \rangle\}}, \quad x, y \in \mathbb{R}^d, \quad (9)$$

where the factor 2 arises from the normalization induced by the Barker balancing function.

Sampling mechanism (auxiliary-variable view). A key practical advantage is that one can sample from $q_B(\cdot | x)$ without needing to evaluate its normalizing constant: the method admits an auxiliary-variable construction (Algorithm 1 in 7) that samples a base increment and then stochastically chooses a direction via logistic probabilities depending on $\nabla \log \pi(x)$. This is the mechanism implemented in the `rmcmc` package [16].

MH correction. Although inspired by Barker’s classical acceptance function [8], the Barker proposal here is used as an MH *proposal kernel* and is still corrected by the standard MH acceptance probability (1). In particular, because q_B is generally asymmetric, the acceptance probability is not $\pi(y)/(\pi(x) + \pi(y))$ (that expression applies to Barker’s accept/reject rule under symmetric proposals). For the Barker proposal, (author?) [7] derive a convenient form:

$$\alpha_B(x, y) = \min \left\{ 1, \frac{\pi(y) (1 + \exp\{\langle \nabla \log \pi(x), (y - x) \rangle\})}{\pi(x) (1 + \exp\{\langle \nabla \log \pi(y), (x - y) \rangle\})} \right\}. \quad (10)$$

The expression (10) is obtained by plugging (9) into (1) and exploiting symmetry of μ_σ .

Adaptation. We adapt scale and shape during warm-up exactly as in Sections 2.2–2.3, then freeze.

2.5 Bimodal Barker (noise-shape modification)

Idea. The Barker proposal depends on a choice of auxiliary “noise” distribution (equivalently, the base increment distribution μ_σ). (author?) [9] study the *optimal design* of the Barker proposal and related locally balanced MH algorithms, and they show that replacing Gaussian auxiliary noise by a carefully chosen *bimodal* noise distribution can improve performance (notably ESJD) on a range of targets.

Bimodal auxiliary noise used in our implementation. We follow the concrete choice implemented as `bimodal_barker_proposal()` in `rmcmc` [16]. Componentwise, the auxiliary noise distribution is an equally weighted mixture of two normals with shared variance σ^2 and means $\pm\sqrt{1-\sigma^2}$:

$$Z_i \sim \frac{1}{2} \mathcal{N}\left(+\sqrt{1-\sigma^2}, \sigma^2\right) + \frac{1}{2} \mathcal{N}\left(-\sqrt{1-\sigma^2}, \sigma^2\right), \quad i = 1, \dots, d, \quad (11)$$

so that $\mathbb{E}[Z_i] = 0$ and $\text{Var}(Z_i) = 1$ for each component. The scalar $\sigma \in (0, 1)$ controls how concentrated each mode is; the default in `rmcmc` is $\sigma = 0.1$ [16]. The Barker construction (proposal kernel plus MH correction) is then applied exactly as in Section 2.4, but with the auxiliary noise distribution replaced by (11).

2.6 Hamiltonian Monte Carlo (HMC): leapfrog dynamics, gradient cost, and Metropolis correction

Let $\pi(x)$ be the target density on $\mathbb{R}^d$ (known up to a normalizing constant). Define the *potential energy*

$$U(x) := -\log \pi(x), \quad \nabla U(x) = -\nabla \log \pi(x),$$

and introduce an auxiliary *momentum* variable $p \in \mathbb{R}^d$ with Gaussian distribution

$$p \sim \mathcal{N}(0, M),$$

where $M \in \mathbb{R}^{d \times d}$ is a symmetric positive definite *mass matrix*. The corresponding Hamiltonian is

$$H(x, p) := U(x) + \frac{1}{2} p^\top M^{-1} p.$$

HMC proposes distant moves by approximately simulating Hamiltonian dynamics using a symplectic integrator (the leapfrog method). Given the current position x_t , we proceed as follows.

Leapfrog integrator. Let $x^{(0)} = x_t$ and sample $p^{(0)} \sim \mathcal{N}(0, M)$. For $\ell = 0, 1, \dots, L-1$, perform one leapfrog step with step size $\epsilon > 0$:

$$p^{(\ell+\frac{1}{2})} = p^{(\ell)} - \frac{\epsilon}{2} \nabla U(x^{(\ell)}), \quad (12)$$

$$x^{(\ell+1)} = x^{(\ell)} + \epsilon M^{-1} p^{(\ell+\frac{1}{2})}, \quad (13)$$

$$p^{(\ell+1)} = p^{(\ell+\frac{1}{2})} - \frac{\epsilon}{2} \nabla U(x^{(\ell+1)}). \quad (14)$$

Denote the numerical endpoint by $(x^*, p^*) := (x^{(L)}, p^{(L)})$. In the standard reversible construction, one additionally negates the momentum, $p^* \leftarrow -p^*$ (this does not change the acceptance probability but ensures reversibility of the proposal mapping).

Gradient evaluation cost. Equations (12)–(14) require evaluating the gradient at

$$\nabla U(x^{(0)}), \nabla U(x^{(1)}), \dots, \nabla U(x^{(L)}),$$

i.e. typically $L+1$ gradient evaluations per HMC proposal trajectory (up to caching/reuse details).

Metropolis correction. Because leapfrog integration is not exact (finite ϵ), HMC applies a Metropolis correction targeting the joint density proportional to $\pi(x) \exp(-\frac{1}{2} p^\top M^{-1} p)$. The proposal is accepted with probability

$$\alpha((x_t, p^{(0)}), (x^*, p^*)) = \min\left\{1, \exp\left(-H(x^*, p^*) + H(x_t, p^{(0)})\right)\right\}.$$

Finally, set

$$x_{t+1} = \begin{cases} x^*, & \text{with probability } \alpha, \\ x_t, & \text{otherwise.} \end{cases}$$

Trajectory length and auxiliary updates (implementation interface). In our implementation, the step size ϵ corresponds to the `scale` parameter, while the choice of trajectory length is controlled either by a fixed L (`n.step`) or by randomizing L via a user-supplied function (`sample.n.step`). Additionally, the auxiliary-variable update `sample.auxiliary` can be used to implement alternatives to full momentum resampling, such as partial momentum refreshment.

3 Evaluation methodology

We evaluate samplers using both *end-of-run diagnostics* (convergence and efficiency summaries) and *running diagnostics* (how quickly posterior moments are learned). Throughout, we run $M \geq 2$ chains per configuration, each with warm-up followed by N retained draws. Let $\theta_t^{(m)}$ denote the (possibly vector-valued) draw at iteration t in chain m after warm-up.

3.1 Basic convergence and efficiency diagnostics

Split $\hat{R}$ (Gelman–Rubin). For a scalar estimand θ , define the per-chain means $\bar{\theta}_m$ and variances s_m^2 :

$$\bar{\theta}_m = \frac{1}{N} \sum_{t=1}^N \theta_t^{(m)}, \quad s_m^2 = \frac{1}{N-1} \sum_{t=1}^N \left(\theta_t^{(m)} - \bar{\theta}_m \right)^2.$$

Let $\bar{\theta} = \frac{1}{M} \sum_{m=1}^M \bar{\theta}_m$ be the overall mean. The within-chain variance and between-chain variance are

$$W = \frac{1}{M} \sum_{m=1}^M s_m^2, \quad B = \frac{N}{M-1} \sum_{m=1}^M (\bar{\theta}_m - \bar{\theta})^2.$$

An over-dispersed estimate of the marginal variance is

$$\hat{V} = \frac{N-1}{N} W + \frac{1}{N} B,$$

and the potential scale reduction factor is

$$\hat{R} = \sqrt{\frac{\hat{V}}{W}}. \quad (15)$$

Values near 1 indicate agreement between and within chains. In practice we use modern rank-normalized split- $\hat{R}$ variants as implemented in contemporary diagnostic toolchains [17].

Effective sample size (ESS). For correlated draws, ESS approximates the number of independent samples that would yield the same Monte Carlo accuracy. For a scalar estimand, a standard definition is

$$\text{ESS} = \frac{MN}{1 + 2 \sum_{k=1}^{\infty} \rho(k)}, \quad (16)$$

where $\rho(k)$ is the lag- k autocorrelation of the (appropriately combined) chain. In practice, the infinite sum is truncated using stable estimators (e.g. initial monotone sequence estimators) and computed separately for bulk and tail functionals [17]. We report:

- *Bulk-ESS*: ESS for rank-normalized draws, reflecting mixing in the central mass.
- *Tail-ESS*: ESS for extreme quantiles (e.g. 5% and 95%), reflecting tail exploration.

Monte Carlo standard error (MCSE). For an estimand $\psi = \mathbb{E}_{\pi}[h(\theta)]$ estimated by $\hat{\psi}$, the MCSE is

$$\text{MCSE}(\hat{\psi}) = \sqrt{\widehat{\text{Var}}(\hat{\psi})}.$$

For the posterior mean of a scalar parameter, a widely used approximation is

$$\text{MCSE}(\bar{\theta}) \approx \frac{\hat{\sigma}_{\theta}}{\sqrt{\text{ESS}}}, \quad (17)$$

where $\hat{\sigma}_{\theta}$ is the estimated marginal posterior standard deviation and ESS is as in (16). We compute MCSEs for means and selected quantiles.

Runtime and ESS/sec. We record wall-clock runtime T per sampler configuration, including gradient evaluations and any warm-up adaptation. To combine statistical and computational efficiency, we report

$$\text{ESS/sec} = \frac{\text{ESS}}{T}.$$

This normalization is essential when comparing gradient-free and gradient-based samplers, as their per-iteration costs can differ by orders of magnitude.

3.2 Running diagnostics (learning speed)

End-of-run diagnostics are necessary but do not capture *transient* behavior: two samplers might have similar final ESS yet converge to correct posterior moments at very different rates. When reference draws are available (PosteriorDB), we compute running-time metrics against “ground truth” moments.

Let $\theta \in \mathbb{R}^d$ denote a vector of parameters of interest in *constrained* coordinates (i.e. the model’s natural parameterization), and let (μ^*, v^*) be reference estimates of its mean and marginal variance vector computed from high-quality reference draws.

Running mean-squared error (MSE) of posterior means. Let $\hat{\mu}_n \in \mathbb{R}^d$ be the pooled running mean after n retained draws per chain (or equivalently after Mn pooled draws). Define

$$\text{MSE}_n = \frac{1}{d} \sum_{j=1}^d (\hat{\mu}_{n,j} - \mu_j^*)^2. \quad (18)$$

A fast sampler should drive MSE_n down quickly as n increases.

Running log-variance distance. Let $\hat{v}_n \in \mathbb{R}^d$ be the running estimate of marginal variances. We define the log-variance discrepancy

$$D_T(n) = \left(\frac{1}{d} \sum_{j=1}^d (\log \hat{v}_{n,j} - \log v_j^*)^2 \right)^{1/2}. \quad (19)$$

Using log stabilizes relative-scale errors: over- and under-estimation by multiplicative factors are penalized symmetrically.

Expected squared jump distance (ESJD). Let $x_t \in \mathbb{R}^d$ denote the unconstrained chain state (the coordinate system in which the Markov kernel operates). The ESJD over N retained draws is

$$\text{ESJD} = \frac{1}{N-1} \sum_{t=1}^{N-1} \|x_{t+1} - x_t\|^2, \quad (20)$$

computed per chain and then averaged across chains. ESJD is an exploration proxy: larger jumps suggest faster movement through state space. However, ESJD must be interpreted together with acceptance probabilities and moment diagnostics, because excessively large proposed jumps may simply be rejected.

4 Benchmark models

4.1 PosteriorDB targets and unconstrained parameterization

We benchmark on posteriors drawn from PosteriorDB, a curated collection of model–data pairs with reference posterior draws intended for testing and comparing inference algorithms [3]. For Stan models, we evaluate $\log \pi(x)$ and $\nabla \log \pi(x)$ in unconstrained coordinates $x \in \mathbb{R}^d$ using BridgeStan [2]. When reporting moment-based running diagnostics, we map unconstrained draws back to constrained coordinates to align with the reference draws.

A benchmark suite should stress qualitatively different failure modes. In this paper we focus on:

- *Funnel geometry and parameterization sensitivity* (Eight Schools, centered vs non-centered).
- *Multimodality* (a low-dimensional Gaussian mixture target).

These are complementary: the former stresses local curvature and strong coupling; the latter stresses global exploration.

4.2 Eight Schools: centered and non-centered

The Eight Schools model is a canonical hierarchical Bayesian benchmark [18, 19]. The data are $y_i \in \mathbb{R}$ with known standard errors $\sigma_i > 0$ for $i = 1, \dots, 8$. The hierarchical model is:

$$y_i \mid \theta_i \sim \mathcal{N}(\theta_i, \sigma_i^2), \quad i = 1, \dots, 8, \quad (21)$$

$$\theta_i \mid \mu, \tau \sim \mathcal{N}(\mu, \tau^2), \quad i = 1, \dots, 8, \quad (22)$$

$$\tau \sim \text{Cauchy}^+(0, 5), \quad (23)$$

with $\mu \in \mathbb{R}$ given the (improper) flat prior in the standard Stan formulation (i.e. no explicit prior term for μ is added). This is exactly the model structure used in the PosteriorDB Stan implementation of the Eight Schools example.

Centered parameterization. The centered form treats $\theta_{1:8}$ as primary parameters and uses (22) directly. The posterior geometry can exhibit a *funnel* structure: when τ is small, the θ_i are tightly concentrated around μ , inducing strong coupling between τ and $\theta_{1:8}$. This produces narrow, curved regions of high probability that are challenging for random-walk proposals and can require careful tuning even for gradient-based methods.

Non-centered parameterization. The non-centered form introduces auxiliary standard normals η_i :

$$\eta_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad \theta_i = \mu + \tau \eta_i. \quad (24)$$

This reparameterization often reduces posterior curvature and decouples τ from the school-level effects when the data are weakly informative, improving mixing dramatically [19]. The comparison of centered vs non-centered is therefore a targeted test of *robustness to geometry* and the effectiveness of covariance adaptation.

4.3 Low-dimensional Gaussian mixture: `low_dim_gauss_mix-low_dim_gauss_mix`

To stress global exploration, we include the PosteriorDB target `low_dim_gauss_mix-low_dim_gauss_mix`, described in PosteriorDB as a *two-dimensional Gaussian mixture model* and represented in a low-dimensional unconstrained parameterization (dimension $d = 5$ in the unconstrained space in the PosteriorDB benchmark listings). Multimodal targets are a known stress test for local gradient methods: gradients tend to pull proposals toward the nearest mode, and energy barriers can lead to mode trapping.

This model is particularly relevant for Barker-type proposals because:

- the Barker construction modifies local proposal directionality without relying on a deterministic drift as in MALA, and
- bimodal auxiliary noise (Section 2.5) is explicitly motivated as a mechanism that can improve exploration and ESJD on representative targets [9].

We therefore use the mixture model to probe whether the relative benefits of robustness-oriented proposals persist beyond unimodal or near-Gaussian regimes.

Scope note

This document defines the samplers, the benchmark targets, and the diagnostic methodology. No numerical results are reported here; results and discussion are presented in later sections of the full paper.

4.4 A sampler-agnostic adaptive warmup wrapper

We implement each sampler through a common interface: for any choice of *scale* $s > 0$ and *shape* matrix $\Sigma \in \mathbb{R}^{d \times d}$ (symmetric positive definite), the sampler induces a Markov transition kernel

$$K_{s, \Sigma}(x, dx'),$$

which leaves π invariant when (s, Σ) are held fixed.¹ During an initial warmup of T_w iterations we adapt (s, Σ) online, then freeze them and sample from the resulting fixed kernel.

¹For HMC, Σ is interpreted as a mass matrix (or its inverse, depending on convention), whereas for Metropolis–Hastings proposals Σ is a proposal covariance/preconditioner.

Algorithm 1 Generic adaptive warmup wrapper for a family of MCMC kernels $\{K_{s,\Sigma}\}$.

Require: Initial state $x_0 \in \mathbb{R}^d$; warmup length T_w ; sampling length T_m ; initial scale $s_0 > 0$ and shape $\Sigma_0 \succ 0$; target acceptance probability δ .

Require: A *scale adapter* **ScaleUpdate** (stochastic approximation or dual averaging) and a *shape adapter* **ShapeUpdate** (diagonal “variance” or full “covariance”).

- 1: Initialize any internal state required by **ScaleUpdate** (e.g. running averages / dual variables).
- 2: **for** $t = 0, 1, \dots, T_w - 1$ **do**
- 3: **(Sampler step)** Draw $(x_{t+1}, \alpha_t) \sim \text{Step}(K_{s_t, \Sigma_t}, x_t)$, where α_t is the acceptance probability used in that step.
- 4: **(Scale adaptation)** Update the scale using the observed acceptance:

$$s_{t+1} \leftarrow \text{ScaleUpdate}(s_t, \alpha_t, t; \delta).$$

- 5: **(Shape adaptation)** Update the shape using the new state:

$$\Sigma_{t+1} \leftarrow \text{ShapeUpdate}(\Sigma_t, x_{t+1}, t).$$

- 6: **end for**
 - 7: **Freeze** (s, Σ) after warmup (either $(s, \Sigma) = (s_{T_w}, \Sigma_{T_w})$ or a smoothed variant recommended by the scale adapter).
 - 8: **for** $t = T_w, \dots, T_w + T_m - 1$ **do**
 - 9: Draw $x_{t+1} \sim K_{s, \Sigma}(x_t, \cdot)$ (no further adaptation).
 - 10: **end for**
 - 11: **return** $\{x_t\}_{t=T_w}^{T_w+T_m}$.
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Notation. Let $\pi(x)$ be the target density on $\mathbb{R}^d$ (known up to normalizing constant), and define $\ell(x) := \log \pi(x)$. When needed, we write $g(x) := \nabla \ell(x)$. Let $\delta \in (0, 1)$ denote the target acceptance probability. Let $\alpha_t \in [0, 1]$ denote the acceptance probability of the t -th proposal produced by K_{s_t, Σ_t} (for HMC, α_t is the Metropolis correction based on the Hamiltonian energy error).

Adaptive wrapper. The warmup phase produces a sequence $\{(x_t, s_t, \Sigma_t)\}_{t=0}^{T_w}$ by alternating: (i) one step of the sampler at current (s_t, Σ_t) , (ii) an update of the scale s_t driven by α_t , and (iii) an update of the shape Σ_t driven by the realized states.

Role of the sampler-specific definitions. Algorithm 1 is agnostic to the choice of sampler. The only sampler-dependent component is the routine $\text{Step}(K_{s, \Sigma}, x)$, which must (i) generate the next state according to the selected kernel and (ii) report the associated acceptance probability α used by that kernel. In subsequent sections we specify $K_{s, \Sigma}$ and α for each sampler considered (AMH, MALA, Barker-type proposals, and HMC).

5 Benchmark results

5.1 Eight Schools (non-centered): comparative sampler behavior

We first benchmark all methods on the non-centered Eight Schools posterior `eight_schools-eight_schools_noncentered` (Section 4.2). This target serves as a “sanity” model: the non-centered reparameterization largely removes the funnel coupling, so differences between samplers are driven primarily by (i) the effectiveness and stability of warmup adaptation, (ii) the ability to learn marginal scales, and (iii) computational cost. Performance is assessed using the running diagnostics defined in Section 3.2: running mean MSE (18), running log-variance distance $D_T(n)$ (19), and ESJD (20). Figure references below correspond to the three dashboard curves (MSE / D_T / ESJD) produced for each batch.

Adaptive random-walk baseline (AMH). Among AMH configurations, diagonal (variance-only) shape adaptation consistently provides the best performance on this target. Both diagonal AMH variants rapidly reach the low plateau in running MSE and attain the smallest $D_T(n)$ values, while maintaining substantially larger ESJD than the corresponding full-covariance AMH variants. In contrast, full covariance adaptation performs poorly in this experiment: the covariance-shaped AMH runs exhibit (i) persistently larger MSE and $D_T(n)$ and (ii) markedly reduced ESJD, indicating that accepted moves are small and exploration is hindered. Operationally, this pattern is consistent with an ill-conditioned or mis-scaled learned covariance (e.g. noisy or unstable full-cov

estimates), which can collapse the effective proposal geometry even when regularization is applied. On this non-centered target, where posterior correlations are comparatively mild, the diagonal adaptation appears to be both sufficient and more robust.

Gradient-based Metropolis proposals (AMALA, Barker, Bimodal Barker). For the non-centered Eight Schools posterior, AMALA, Barker, and Bimodal Barker achieve similar end-state running MSE, suggesting that mean estimation is not a separating stressor once geometry is regularized by the parameterization. Differences are more visible in variance learning and exploration proxies: AMALA and Bimodal Barker reach smaller $D_T(n)$ values than standard Barker, and their ESJD traces are larger, indicating more aggressive movement in unconstrained space. The bimodal auxiliary noise improves over standard Barker in this experiment, consistent with its design goal of increasing typical jump distances while retaining MH correction. We note, however, that these gradient-MH results were obtained with a single chain in this batch, so they should be interpreted as within-chain performance signals rather than definitive multi-chain convergence evidence.

Hamiltonian Monte Carlo (HMC): trajectory length sensitivity and cost. HMC performance is strongly sensitive to the trajectory length parameter L in this benchmark. With short trajectories (fixed $L = 10$), ESJD is extremely small and both running MSE and $D_T(n)$ decay more slowly, indicating that the method behaves like an overly conservative local sampler. Increasing the trajectory length (fixed $L = 50$) substantially improves exploration and accelerates convergence of both means and variances. An adaptive step-count variant with auxiliary refresh further increases ESJD, but delivers only marginal additional improvement in running MSE/ $D_T(n)$ on this already-regular target. Because each HMC proposal requires $\mathcal{O}(L)$ gradient evaluations, these gains must be interpreted together with runtime: on the non-centered Eight Schools posterior, longer trajectories increase statistical efficiency per iteration but can be dominated by gradient cost in ESS/sec comparisons. (In our implementation, HMC was run with conservative, stability-oriented settings; in particular, step-size adaptation was disabled when required to avoid gradient overflows.)

Summary and practical takeaways (this model). On the non-centered Eight Schools target, most samplers reach comparable accuracy in posterior means, so the most informative separation comes from variance learning (D_T), exploration (ESJD), and computational efficiency. Diagonal AMH provides a strong gradient-free baseline with stable adaptation and good moment learning. AMALA and Bimodal Barker exhibit stronger exploration (higher ESJD) and good variance learning, while standard Barker is comparatively weaker in this batch. HMC requires sufficiently long (or randomized) trajectories to behave as an effective global sampler; however, on this target the additional compute from large L can yield limited practical benefit beyond the point where running MSE and D_T have already stabilized.

5.2 Low-dimensional Gaussian mixture (low_dim_gauss_mix): multimodality, mode trapping, and why ESJD/ESS can mislead

This target is explicitly *multimodal*. The running diagnostics in Section 3.2 are computed against PosteriorDB reference moments, i.e. the *global* mixture mean and variances. Consequently, any sampler that remains confined to a single mode will typically exhibit (i) a plateau in the running mean error MSE_n (18) and (ii) a plateau in the running log-variance discrepancy $D_T(n)$ (19), even if within-mode mixing is excellent.

AMH (four configurations): diagonal shape succeeds; full covariance can fail on multimodality. Figures ??-?? compare the four AMH variants (stochastic-approximation vs dual-averaging scale adaptation; diagonal “variance” vs full “covariance” shape). Two robust patterns emerge:

1. **Diagonal shape (variance) learns the global moments quickly.** Both variance-shape runs drive MSE_n and $D_T(n)$ essentially to zero over the main sampling window, indicating that the empirical means and marginal variances match the reference moments.
2. **Full covariance shape can be actively harmful on a mixture.** Both covariance-shape runs plateau at substantially larger MSE_n and $D_T(n)$ (and also exhibit poorer end-of-run mixing diagnostics relative to the diagonal runs), indicating that the chains are not reproducing the global mixture moments. A plausible mechanism is that covariance adaptation is *mode-local*: if warmup samples concentrate in one mode, the empirical covariance largely reflects within-mode correlations and scales. The resulting proposal can become *well-tuned for local exploration but poorly aligned with the inter-mode direction*, reducing the probability of barrier-crossing moves needed for correct global averaging.

A key cautionary example appears in the ESJD curves: the dual-averaging + covariance AMH configuration attains the *largest* ESJD among AMH runs, yet it is the *worst* in MSE_n and $D_T(n)$. This illustrates that on

multimodal targets, large average jumps can still correspond to rapid *within-mode* movement rather than genuine *between-mode* exploration.

Gradient-based MH (MALA, Barker, Bimodal Barker): strong local movement, wrong global moments under single-chain initialization. Figures ??–?? show that MALA, Barker, and Bimodal Barker achieve high and stable ESJD values (MALA highest, then Bimodal Barker, then Barker), consistent with efficient local exploration once warmup has stabilized scale/shape. However, all three methods exhibit essentially the *same non-vanishing plateaus* in MSE_n and $D_T(n)$ (at values far from zero), which is the diagnostic signature of *mode trapping*: the chain(s) explore one mode thoroughly but do not traverse the low-density region separating modes often enough to recover the global mixture moments.

This behavior is expected for locally informed proposals on multimodal landscapes. In MALA, the drift term $\Sigma \nabla \log \pi(x)$ points toward increasing log density, i.e. back into the current basin. For Barker-type proposals, the logistic tilting in (9) similarly biases directions according to the local gradient. Such mechanisms improve robustness and efficiency *within* a mode, but they do not by themselves create a global transition mechanism for crossing energy barriers. The bimodal auxiliary noise is designed to improve jump behavior in locally balanced schemes, but in this particular run it did not materially reduce the global-moment error, suggesting that the remaining limitation is inter-mode communication rather than within-mode step length. We emphasize that these BBA results are based on a single chain in this batch; they should therefore be read as within-chain performance signals rather than definitive multi-chain convergence evidence.

HMC (trajectory-length sensitivity): larger ESJD with longer/adaptive paths, but still global-moment plateaus. Figures ??–?? compare HMC with (i) fixed $L = 10$ steps, (ii) fixed $L = 50$ steps, and (iii) an adaptive step-count variant with auxiliary refresh. As expected, increasing trajectory length (and/or randomizing it) substantially increases ESJD and improves *local* exploration: $L = 10$ yields the smallest ESJD and the worst MSE_n , whereas $L = 50$ and the adaptive variant achieve much larger ESJD and smaller (but still nonzero) MSE_n .

Crucially, *none* of the HMC variants drives MSE_n or $D_T(n)$ toward zero here: both running diagnostics plateau at levels comparable to the gradient-based MH runs. This is consistent with a standard multimodality failure mode for HMC: Hamiltonian trajectories approximately conserve energy, so moving between modes typically requires crossing a low-probability (high potential energy) barrier. Longer trajectories can improve exploration *within* the current energy basin, but do not guarantee inter-basin transitions unless the momentum/step-size regime allows barrier crossing with non-negligible probability.

Implementation constraint for HMC on low_dim_gauss_mix: conservative step size. For this mixture target, HMC was run with a deliberately conservative step size (in our interface, the parameter `scale`, i.e. the leapfrog step size ϵ in (12)–(14)). Empirically, using step sizes larger than $\epsilon \approx 10^{-2}$ caused the quantity returned by the model gradient routine (the `log_gradient_density`) to overflow (numerically producing $\pm\infty$ or NaN) at intermediate leapfrog states. We therefore fixed $\epsilon = 0.01$ for HMC on this model to ensure numerical stability.

This behavior is expected in principle because the leapfrog integrator evaluates gradients at a sequence of intermediate positions $x^{(0)}, x^{(1)}, \dots, x^{(L)}$ along the simulated trajectory. If ϵ is too large, then a single update

$$x^{(\ell+1)} = x^{(\ell)} + \epsilon M^{-1} p^{(\ell+\frac{1}{2})}$$

can move the state into regions where the log density (and especially its gradient) has extremely large magnitude. For mixture posteriors, such regions can arise when (i) one component has very small effective scale in some direction, producing sharp curvature (large Hessian eigenvalues), or (ii) the trajectory crosses low-density barrier regions where the posterior is effectively dominated by a single component and gradient terms can become large in absolute value. When this happens, evaluating $\nabla \log \pi(x)$ can exceed floating-point range, and the resulting Hamiltonian update becomes undefined.

To make this concrete, we inspected a representative call to the target’s log-density-and-gradient routine in unconstrained coordinates $x \in \mathbb{R}^5$. At that point, the routine returned a log density of

$$\log \pi(x) = -9.95694176768 \times 10^{11},$$

together with the gradient components

$$\nabla \log \pi(x) = \begin{pmatrix} -7.055733 \times 10^5 \\ -1.991390 \times 10^{12} \\ -1.003371 \times 10^3 \\ -6.600792 \times 10^1 \\ 3.418176 \times 10^2 \end{pmatrix}. \quad (25)$$

The presence of a component on the order of 10^{12} illustrates the extreme local sensitivity of the log density in some directions: even a moderate leapfrog displacement Δx can produce a change $\Delta \log \pi(x) \approx \langle \nabla \log \pi(x), \Delta x \rangle$ that is far outside the safe numerical range. Accordingly, larger step sizes ϵ can push intermediate leapfrog states into regimes where either $\log \pi(x)$ or $\nabla \log \pi(x)$ becomes non-finite, triggering the observed `log-gradient-density` overflows. The small step size used here therefore trades computational efficiency for robustness: decreasing ϵ reduces per-step integration error and helps keep the trajectory within numerically safe regions, but it also limits the distance traveled per gradient evaluation, potentially requiring larger L to achieve comparable exploration.

Takeaway for this benchmark: report a mode-switching diagnostic (not only ESS/ $\hat{R}$). Across HMC and the gradient-based MH methods, end-of-run quantities like ESS (and even $\hat{R}$ when only one chain is available) can look satisfactory while MSE_n and $D_T(n)$ clearly indicate failure to match the *global* reference distribution. For multimodal targets, it is therefore essential to pair standard diagnostics with a simple *mode occupancy / switching* statistic (e.g. the fraction of draws with $x_1 > 0$ and the number of sign changes of x_1), and to use overdispersed initializations that deliberately place chains in different basins when possible.

5.3 sblrc-blr: correlated posterior, adaptation pathologies, and gradient-based efficiency

We next benchmark all methods on the PosteriorDB target `sblrc-blr`, which serves as an “applied” posterior with nontrivial correlation structure in unconstrained coordinates. Performance is assessed using the running diagnostics from Section 3.2: running mean error MSE_n (18), running log-variance distance $D_T(n)$ (19), and ESJD (20). Figure references below correspond to the three dashboard curves ($\text{MSE} / D_T / \text{ESJD}$) produced for each batch.

Adaptive random-walk Metropolis (AMH): diagonal adaptation is robust; full covariance adaptation collapses. Figures ??–?? compare four AMH configurations (stochastic approximation vs dual averaging for scale adaptation; diagonal “variance” vs full “covariance” for shape). Two clear patterns emerge. First, *diagonal (variance) shape* is consistently superior: both variance-shaped AMH runs reduce MSE_n and $D_T(n)$ substantially faster and to markedly lower plateaus than the covariance-shaped variants, and they maintain larger ESJD, indicating more effective exploration per accepted move. Second, *full covariance adaptation fails catastrophically* in this experiment: the covariance-shaped runs plateau at much larger MSE_n and $D_T(n)$ and exhibit very small ESJD, consistent with a mis-scaled or ill-conditioned learned covariance that effectively “freezes” the random walk. A plausible mechanism is that the empirical covariance estimate during warmup is unstable in these coordinates (e.g. noisy cross-covariances, strong anisotropy, or near-singularity after regularization), so the resulting proposal geometry becomes overly conservative in some directions and poorly aligned in others. The diagonal estimate, while less expressive, appears far more robust here.

Gradient-based MH (AMALA, Barker, Bimodal Barker): rapid moment learning and strong efficiency. Figures ??–?? show that AMALA, Barker, and Bimodal Barker all drive MSE_n and $D_T(n)$ essentially to zero over the sampling window, indicating that both posterior means and marginal variances match the PosteriorDB reference moments for this target. Their ESJD traces separate in the expected direction: AMALA attains the largest ESJD, followed by Bimodal Barker and then standard Barker. This ordering is consistent with the underlying proposal mechanisms: AMALA uses a deterministic drift aligned with $\nabla \log \pi(x)$ (Section 2.3), while Barker-type proposals tilt local moves stochastically via the balancing construction (Section 2.4), with bimodal auxiliary noise designed to increase typical jump distances (Section 2.5). Operationally, the key result for `sblrc-blr` is that gradient-informed MH proposals provide both accurate moment recovery and substantially stronger exploration than the random-walk baseline.

HMC: trajectory-length sensitivity under a stability-limited step size. Figures ??–?? compare three HMC variants: fixed $L = 10$, fixed $L = 50$, and an adaptive step-count variant with auxiliary refresh. With short trajectories ($L = 10$), ESJD is extremely small and $D_T(n)$ fails to decay to zero, indicating that the method behaves like an overly local integrator that does not traverse the posterior efficiently under the chosen stability settings. Increasing the trajectory length to $L = 50$ yields a clear improvement: ESJD increases by an order of magnitude and both MSE_n and $D_T(n)$ decay rapidly to (approximately) zero. The adaptive variant with auxiliary refresh also eventually reaches near-zero MSE_n and $D_T(n)$, but it does so more slowly than the fixed $L = 50$ run in this batch. Because each HMC proposal costs $\mathcal{O}(L)$ gradient evaluations, the gains from larger L must be interpreted together with runtime: here $L = 50$ is the configuration that achieves reliable moment recovery, but its improved statistical efficiency per iteration comes with a substantial compute increase.

Implementation constraint for HMC on sblrc-blr: very small step size ($\epsilon = 10^{-4}$). For this target, HMC was run with a deliberately conservative step size (in our interface, `scale`, i.e. the leapfrog step size ϵ in (12)–(14)). Empirically, larger values led to unstable leapfrog trajectories (large energy errors and/or non-finite evaluations at intermediate states), so we fixed $\epsilon = 10^{-4}$ as the most reliable setting across HMC variants. This constraint is consistent with the structure of the leapfrog updates: the momentum half-step $p^{(\ell+\frac{1}{2})} = p^{(\ell)} - \frac{\epsilon}{2} \nabla U(x^{(\ell)})$ scales linearly with both ϵ and the gradient magnitude, and repeated steps can amplify numerical error if ϵ is not small relative to the curvature scale of the target.

To make this concrete, we inspected a representative call to the target’s log-density-and-gradient routine in unconstrained coordinates. At that point, the routine returned

$$\log \pi(x) = -377.2796,$$

with gradient

$$\nabla \log \pi(x) = \begin{pmatrix} 7.026848 \\ 4.063391 \\ 9.004077 \\ -4.930682 \\ -13.399368 \\ -100.610578 \end{pmatrix}. \quad (26)$$

Even with moderate gradient magnitudes (here on the order of 10^2 in the largest component), the leapfrog scheme evaluates $\nabla \log \pi(\cdot)$ at *intermediate* states along the trajectory. If ϵ is too large, a single position update $x^{(\ell+1)} = x^{(\ell)} + \epsilon M^{-1} p^{(\ell+\frac{1}{2})}$ can move into regions where the log density curvature is much sharper than at the initial state, causing rapid growth in energy error and potential numerical instability. Fixing $\epsilon = 10^{-4}$ therefore trades efficiency for robustness: it reduces integration error and helps keep trajectories in numerically safe regions, but it also makes per-step movement small, so sufficiently large L (here $L = 50$) is required to obtain meaningful exploration.

Summary and practical takeaways (this model). On `sblrc-blr`, the most informative separation is between (i) robust, accurate gradient-based MH proposals and (ii) adaptation-sensitive random-walk proposals and stability-limited HMC settings. AMALA, Barker, and Bimodal Barker reliably recover reference means and variances (near-zero MSE_n and D_T) with comparatively strong ESJD, making them the best-performing family in this batch. AMH with diagonal shape adaptation is the only random-walk configuration that behaves stably, but it remains clearly less effective than the gradient-informed methods; full covariance adaptation is detrimental here, consistent with an ill-conditioned learned proposal shape. For HMC, reliable performance is achieved only when trajectory length is sufficiently large (fixed $L = 50$), but the need for a very small step size ($\epsilon = 10^{-4}$) reduces ESJD per gradient evaluation and makes runtime/efficiency comparisons sensitive to the gradient cost budget.

References

- [1] C. Andrieu and J. Thoms. A tutorial on adaptive MCMC. *Statistics and Computing*, 18(4):343–373, 2008.
- [2] E. Roualdes et al. BridgeStan: Efficient in-memory access to Stan log densities and gradients. 2023.
- [3] M. Magnusson et al. PosteriorDB: A database of Bayesian posteriors with reference draws. 2024.
- [4] G. O. Roberts, A. Gelman, and W. R. Gilks. Weak convergence and optimal scaling of random walk Metropolis algorithms. *The Annals of Applied Probability*, 7(1):110–120, 1997.
- [5] G. O. Roberts and J. S. Rosenthal. Optimal scaling of discrete approximations to Langevin diffusions. *Journal of the Royal Statistical Society: Series B*, 60(1):255–268, 1998.
- [6] A. Beskos et al. Optimal tuning of the hybrid Monte Carlo algorithm. *Bernoulli*, 19(5A):1501–1534, 2013.
- [7] S. Livingstone and G. Zanella. The Barker proposal: combining robustness and efficiency in MCMC. *Journal of the Royal Statistical Society: Series B*, 2022.
- [8] A. A. Barker. Monte Carlo calculations of the radial distribution functions for a proton–electron plasma. *Australian Journal of Physics*, 18:119–134, 1965.
- [9] J. Vogrinc, S. Livingstone, and G. Zanella. Optimal design of locally balanced proposals and the bimodal Barker construction. 2023.

- [10] N. Metropolis et al. Equation of state calculations by fast computing machines. *The Journal of Chemical Physics*, 21(6):1087–1092, 1953.
- [11] W. K. Hastings. Monte Carlo sampling methods using Markov chains and their applications. *Biometrika*, 57(1):97–109, 1970.
- [12] L. Tierney. A note on Metropolis–Hastings kernels for general state spaces. *The Annals of Applied Probability*, 8(1):1–9, 1998.
- [13] H. Haario, E. Saksman, and J. Tamminen. An adaptive Metropolis algorithm. *Bernoulli*, 7(2):223–242, 2001.
- [14] Y. F. Atchadé. An adaptive version for the Metropolis adjusted Langevin algorithm with a truncated drift. *Methodology and Computing in Applied Probability*, 8:235–254, 2006.
- [15] G. O. Roberts and R. L. Tweedie. Exponential convergence of Langevin distributions and their discrete approximations. *Bernoulli*, 2(4):341–363, 1996.
- [16] `rmcmc` package documentation. *CRAN*, accessed 2025.
- [17] A. Vehtari, A. Gelman, D. Simpson, B. Carpenter, and P.-C. Bürkner. Rank-normalization, folding, and localization: improved $\hat{R}$ for assessing convergence of MCMC. *Bayesian Analysis*, 16(2):667–718, 2021.
- [18] A. Gelman et al. *Bayesian Data Analysis*. Chapman & Hall/CRC, 3rd edition, 2013.
- [19] O. Papaspiliopoulos, G. O. Roberts, and M. Sköld. A general framework for the parametrization of hierarchical models. *Statistical Science*, 22(1):59–73, 2007.